KB금융그룹 | 국민의 평생 금융파트너

K-디지털 트레이닝

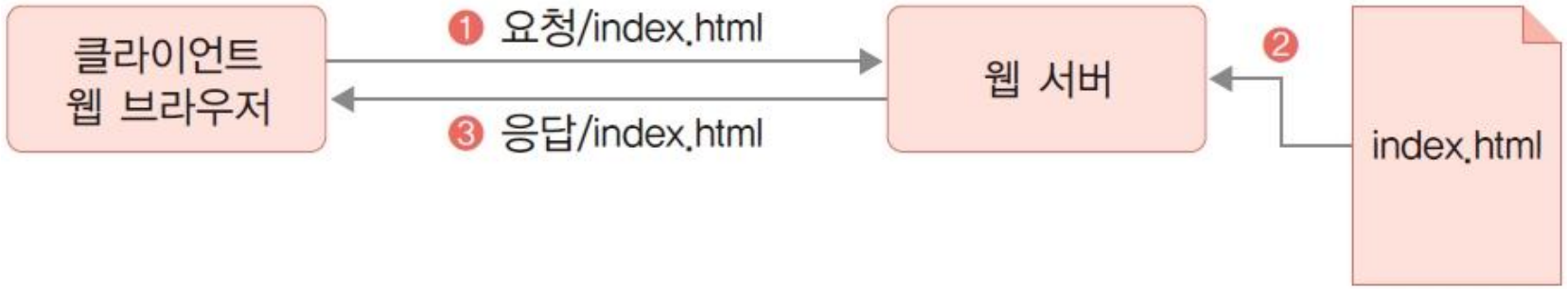
서블릿의 이해

[KB] IT's Your Life

KB 국민은행

서블릿(Servlet) 개요

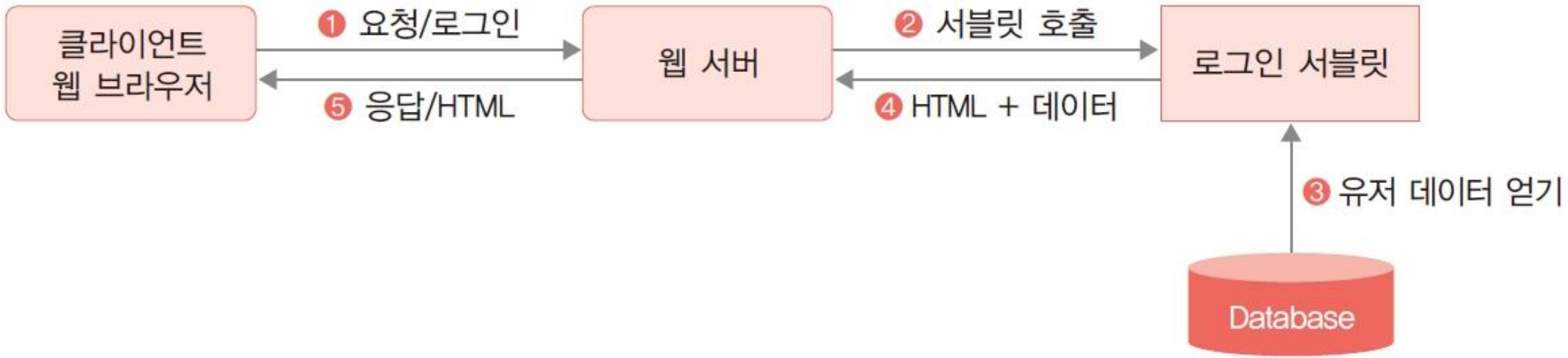
기본적인 웹의 요청과 응답 과정



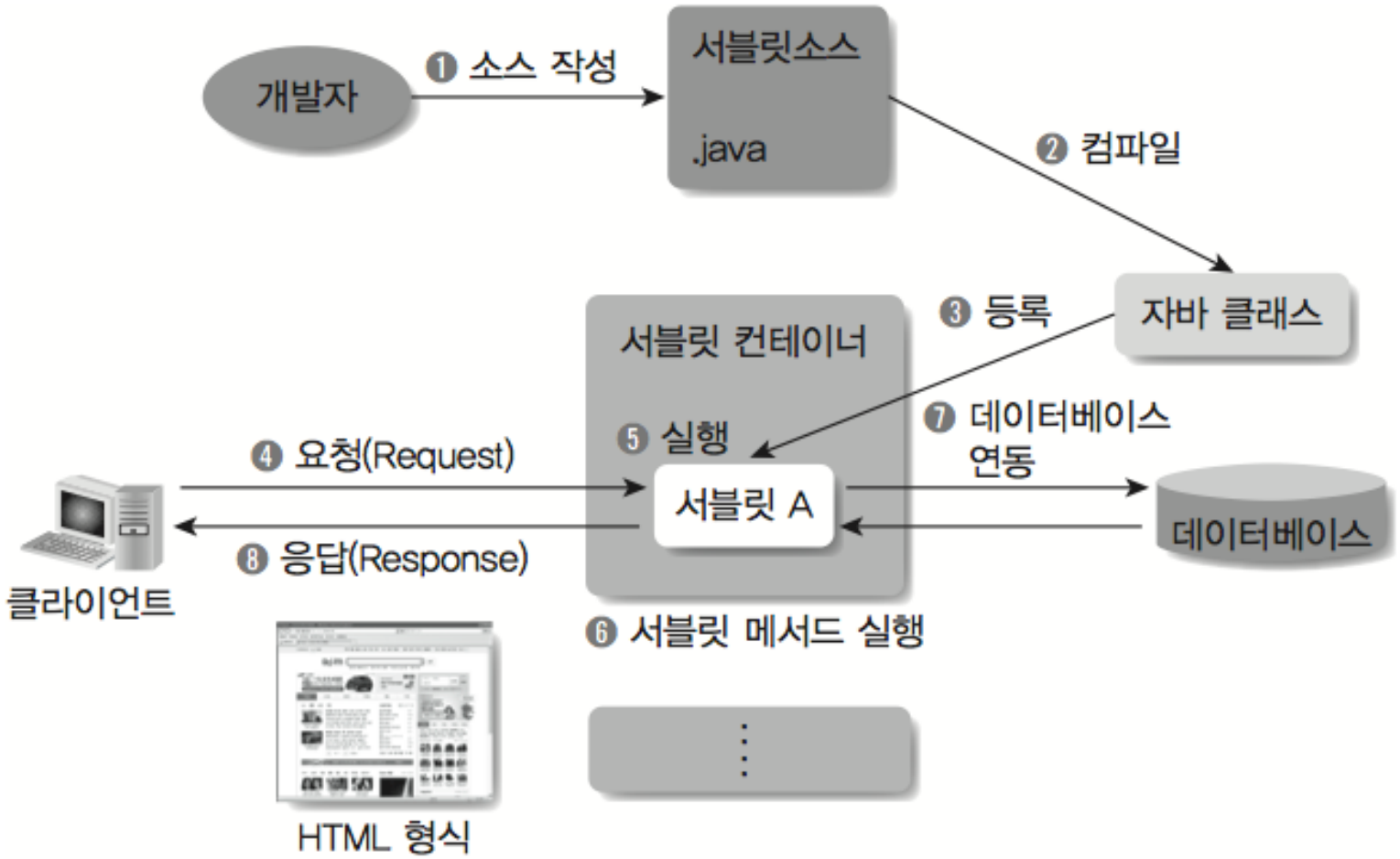
서블릿(Servlet) 개요

서블릿

- 웹 컨테이너에 의해 관리
- 다양한 클라이언트 요청에 의해 컨텐츠로 응답 가능한 자바 기반의 웹 컴포넌트



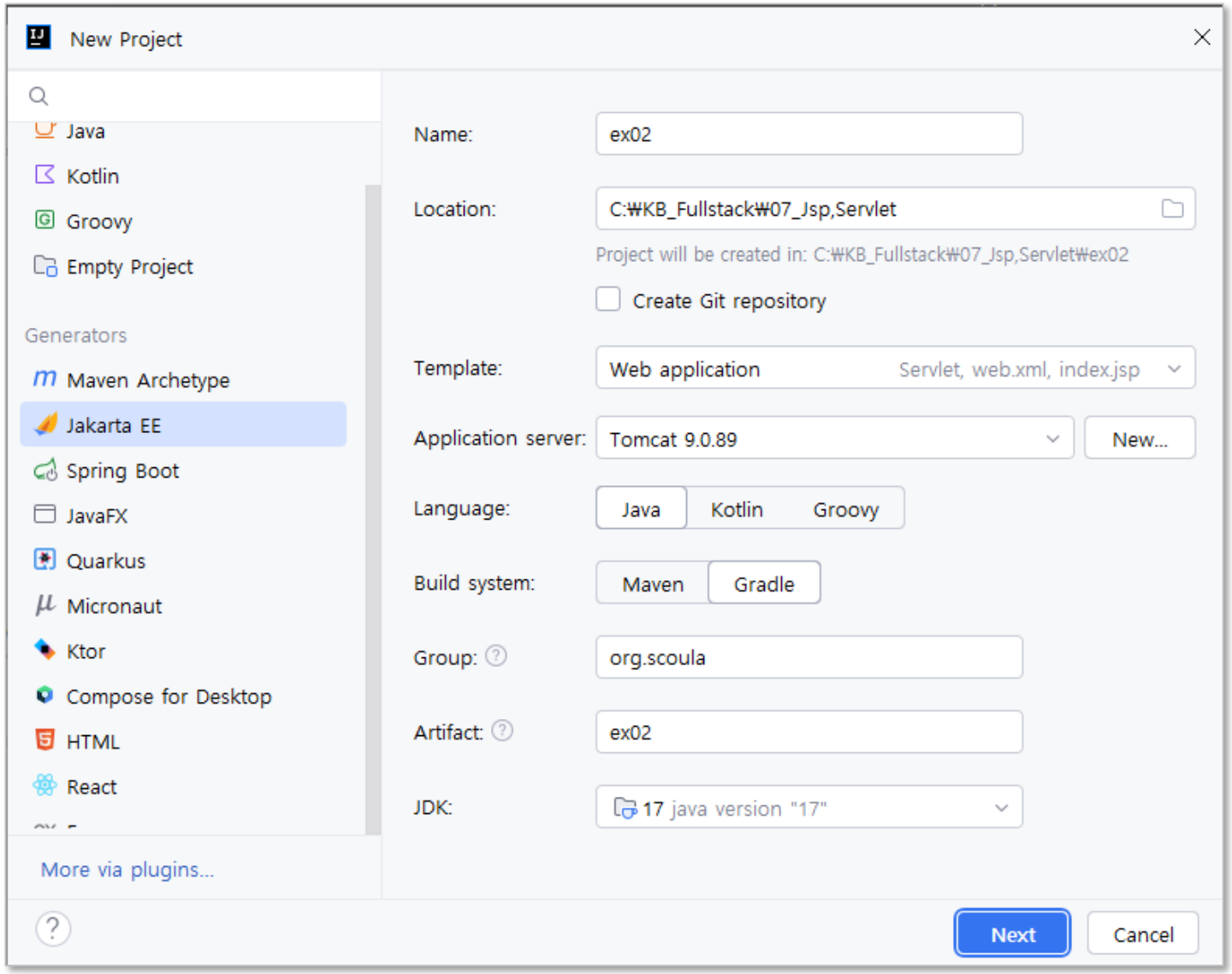
서블릿(Servlet) 개요



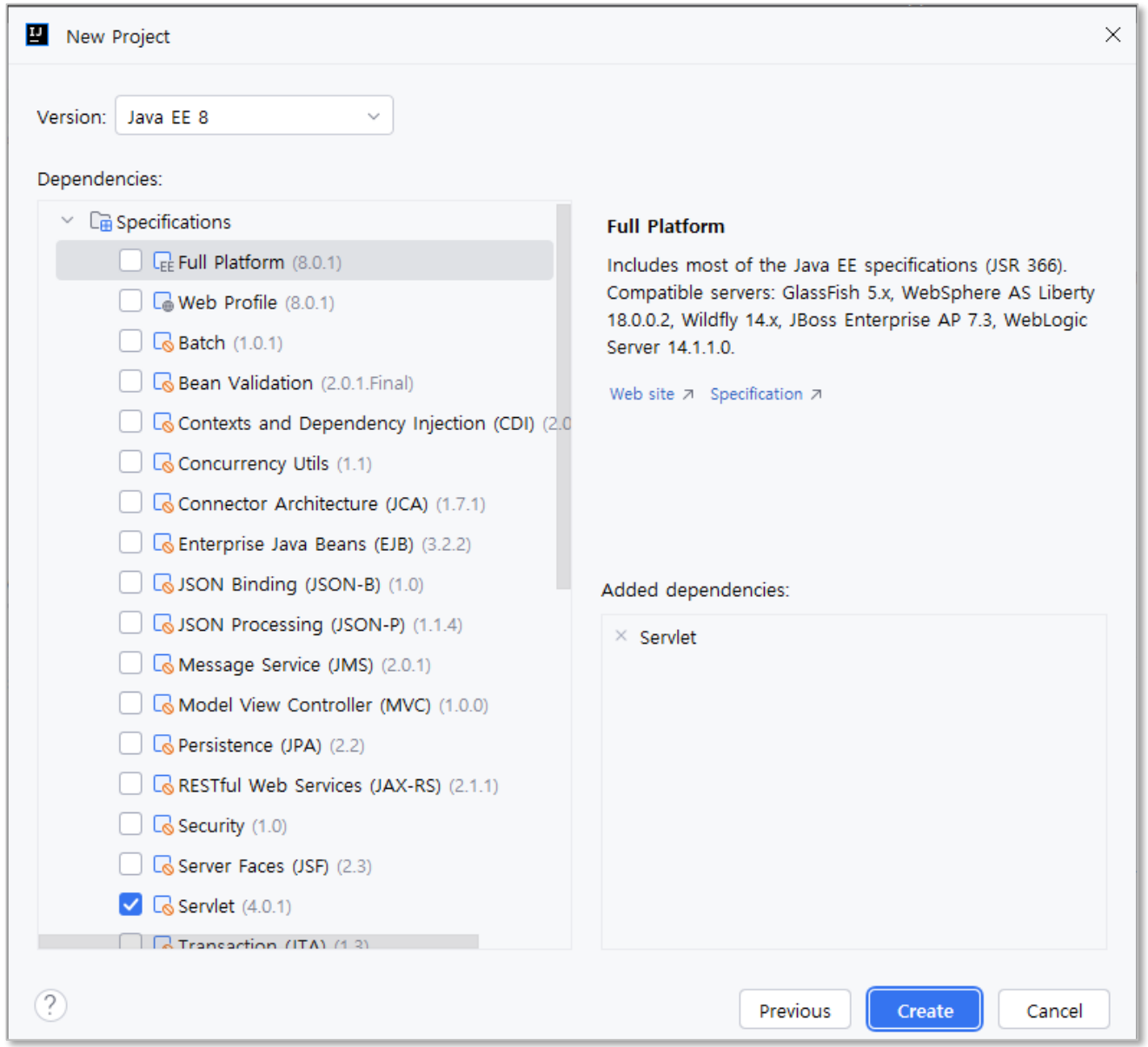
프로젝트 만들기

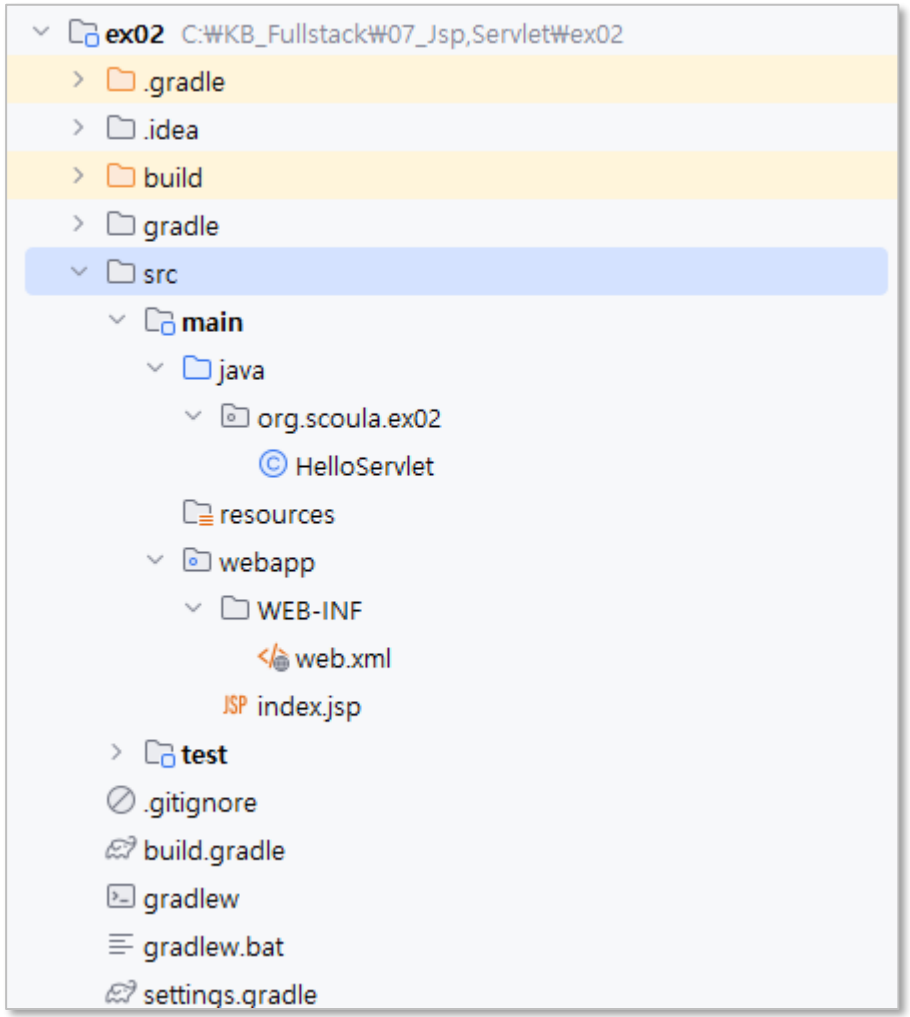
프로젝트 생성

- Templates: Jakarta EE
- Project name: ex02



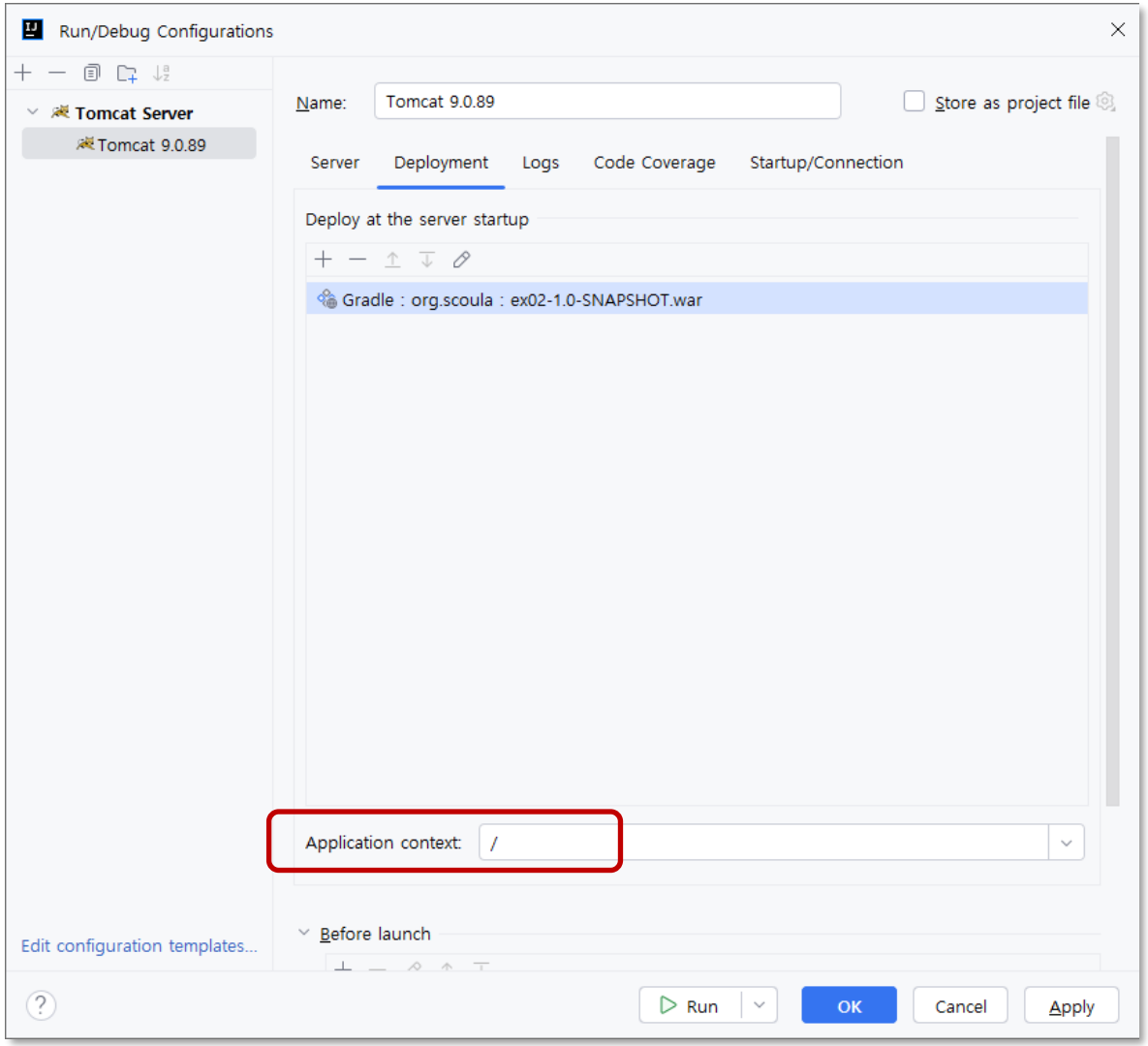
프로젝트 만들기

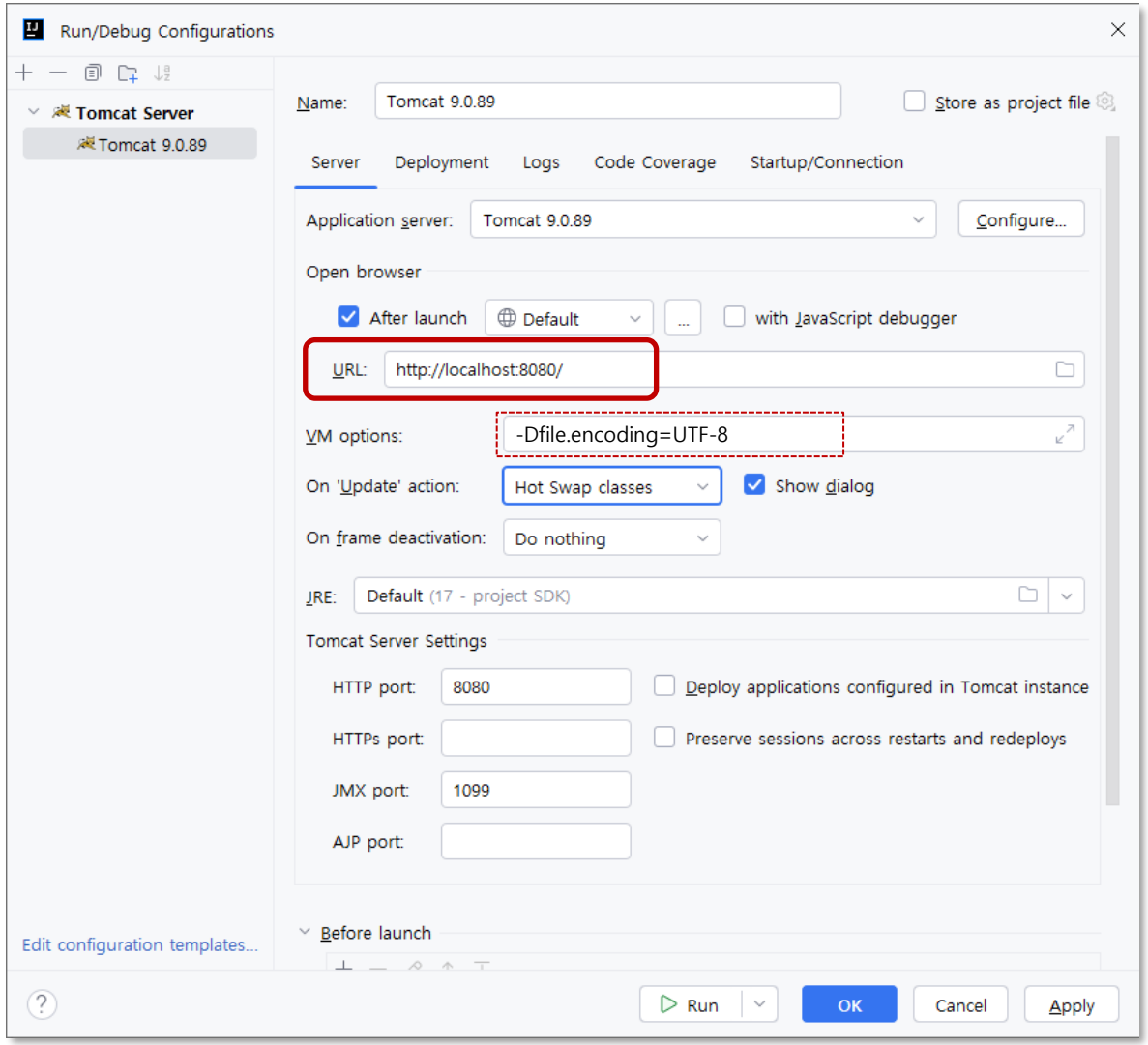

New Project
Version: Java EE 8
Dependencies:
Specifications
Full Platform (8.0.1)
Web Profile (8.0.1)
Batch (1.0.1)
Bean Validation (2.0.1.Final)
Contexts and Dependency Injection (CDI)
Concurrency Utils (1.1)
Connector Architecture (JCA) (1.7.1)
Enterprise Java Beans (EJB) (3.2.2)
JSON Binding (JSON-B) (1.0)
JSON Processing (JSON-P) (1.1.4)
Message Service (JMS) (2.0.1)
Model View Controller (MVC) (1.0.0)
Persistence (JPA) (2.2)
RESTful Web Services (JAX-RS) (2.1.1)
Security (1.0)
Server Faces (JSF) (2.3)
Servlet (4.0.1)
Full Platform
Includes most of the Java EE specifications (JSR 366). Compatible servers: GlassFish 5.x, WebSphere AS Liberty 18.0.0.2, Wildfly 14.x, JBoss Enterprise AP 7.3, WebLogic Server 14.1.1.0.
Web site
Specification
Added dependencies:
Servlet
Previous
Create
Cancel


ex02 C:₩KB_Fullstack₩07_Jsp,Servlet₩ex02
.gradle
.idea
build
gradle
src
main
java
org.scoula.ex02
HelloServlet
resources
webapp
WEB-INF
web.xml
index.jsp
test
.gitignore
build.gradle
gradlew
gradlew.bat
settings.gradle

프로젝트 만들기

Tomcat 설정



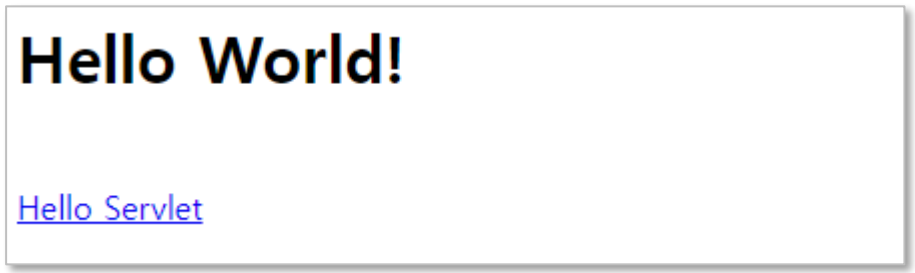


프로젝트 만들기

index.jsp

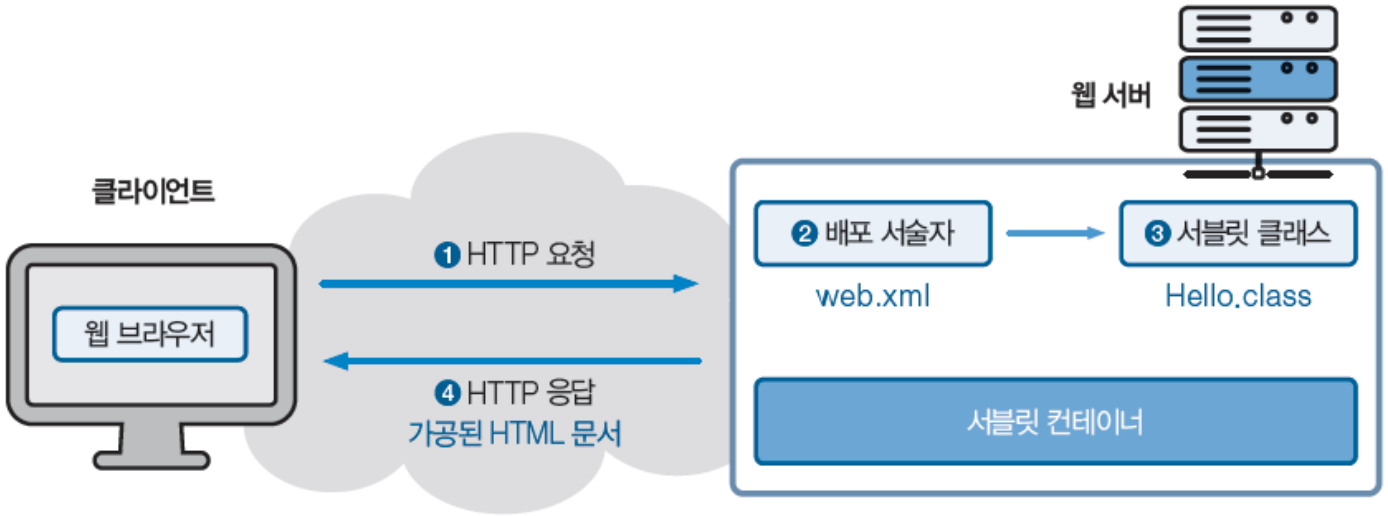
```
<%@ page contentType="text/html; charset=UTF-8" pageEncoding="UTF-8" %>
<!DOCTYPE html>
<html>
<head>
    <title>JSP - Hello World</title>
</head>
<body>
<h1><%= "Hello World!" %>
</h1>
<br/>
<a href="hello-servlet">Hello Servlet</a>
</body>
</html>
```

http://localhost:8080/



서블릿 맵핑

서블릿 동작 과정



HelloServlet.java

```
package org.scoula.ex02;

import java.io.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;

// @WebServlet(name = "helloServlet", value = "/hello-servlet")
public class HelloServlet extends HttpServlet {
    private String message;

    public void init() {
        message = "Hello World!";
    }

    public void doGet(HttpServletRequest req, HttpServletResponse resp) throws IOException, ServletException {
        resp.setContentType("text/html");

        PrintWriter out = resp.getWriter();
        out.println("<html><body>");
        out.println("<h1>" + message + "</h1>");
        out.println("</body></html>");
    }

    public void destroy() {
    }
}
```

서블릿 맵핑

web.xml

```
〈servlet〉
        〈servlet-name〉서블릿별명〈/servlet-name〉
        〈servlet-class〉패키지를 포함한 서블릿명〈/servlet-class〉
〈/servlet〉
〈servlet-mapping〉
        〈servlet-name〉서블릿별명〈/servlet-name〉
        〈url-pattern〉/맵핑명〈/url-pattern〉
〈/servlet-mapping〉
```

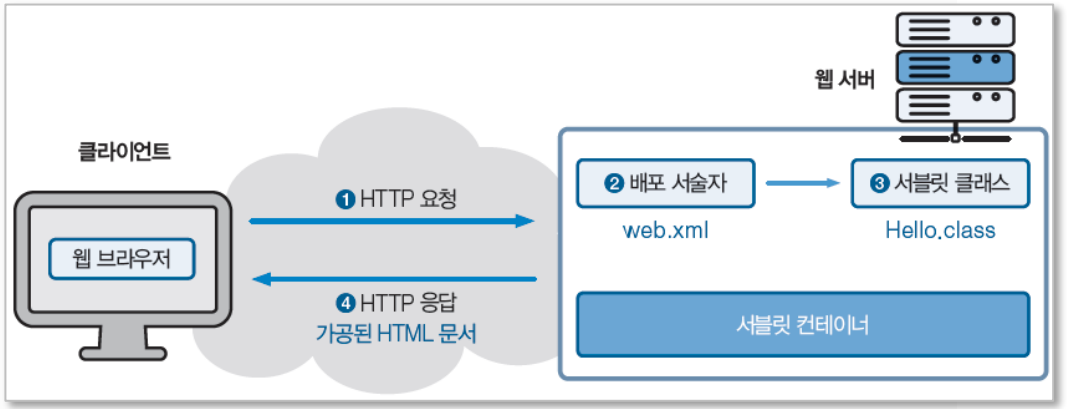
```
package org.scoula.ex02;

…

// @WebServlet(name = "helloServlet", value = "/hello-servlet")
public class HelloServlet extends HttpServlet {
    …
}
```

서블릿 맵핑

web.xml

- web.xml 수정본을 반영하려면 웹서버 재기동

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xmlns="http://xmlns.jcp.org/xml/ns/javaee"
         xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
         id="WebApp_ID" version="3.1">

    <servlet>
        <servlet-name>helloServlet</servlet-name>
        <servlet-class>org.scoula.ex02.HelloServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>helloServlet</servlet-name>
        <url-pattern>/hello-servlet</url-pattern>
    </servlet-mapping>

</web-app>
```



http://localhost:8080/

Hello World!

Hello Servlet

http://localhost:8080/**hello-servlet**

Hello World!

@WebServlet

@WebServlet 어노테이션 이용 방법

- Xml 파일 대신 자바 코드에 기술
- Servlet 3.0의 어노테이션

- @WebServlet
- @WebFilter
- @WebInitParam
- @WebListener
- @MultipartConfig
- @DeclareRoles
- @EJB
- @EJBs
- @Resource
- @Resources
- @PersistenceContext
- @PersistenceContexts
- @PersistenceUnit
- @PersistenceUnits
- @PostConstruct
- @PreDestroy
- @RunAs

@WebServlet

서블릿 맵핑 설정

```
@WebServlet("/맵핑명")
public class MyServlet extends HttpServlet { … }
```

서블릿 별명과 urlPatterns 속성으로 여러 개의 맵핑명 지정

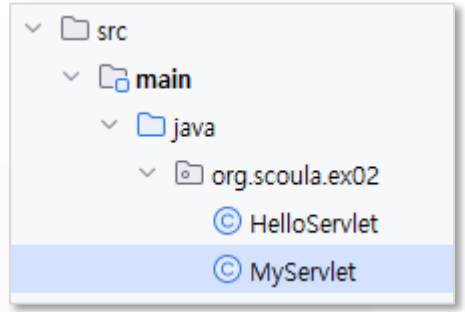
```
@WebServlet(name="서블릿별명",
            urlPatterns={"/맵핑명", "/맵핑명2"})
public class MyServlet extends HttpServlet { … }
```

또는

```
@WebServlet(name="서블릿별명",
            value={"/맵핑명", "/맵핑명2"})
public class MyServlet extends HttpServlet { … }
```

@WebServlet

MyServlet.java 어노테이션을 이용한 서블릿 맵핑 등록



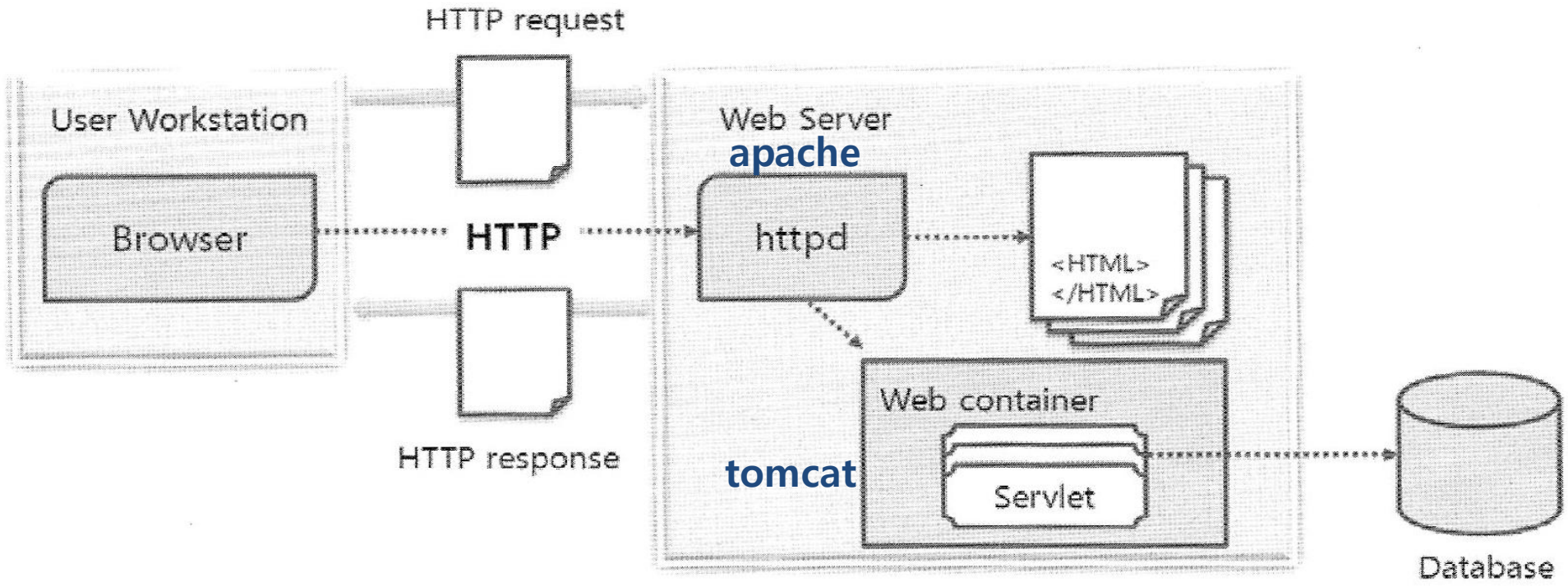

```
@WebServlet(name="MyServlet", urlPatterns={"/xxx", "/yyy" })
public class MyServlet extends HttpServlet {

  protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws IOException, ServletException {

    System.out.println("HelloServlet 요청");
    PrintWriter out = resp.getWriter();
    out.println("<h1>Hello Servlet</h1>");

  }
}
```

- http://localhost/xxx
- http://localhost/yyy


```
서버에 연결됨
[2025-01-10 03:39:23,532] 아티팩트 Gradle : org.scoula : ex0
[2025-01-10 03:39:24,082] 아티팩트 Gradle : org.scoula : ex0
[2025-01-10 03:39:24,082] 아티팩트 Gradle : org.scoula : ex0
10-Jan-2025 15:39:33.271 INFO [Catalina-utility-2] org.apac
10-Jan-2025 15:39:33.350 INFO [Catalina-utility-2] org.apac
HelloServlet 요청
```

서블릿 아키텍처 및 핵심 API

서블릿을 사용하는 경우 웹 아키텍처



서블릿 아키텍처 및 핵심 API

웹브라우저의 URL 요청 → 해당 서블릿 호출

- 웹 컨테이너에서 서블릿을 실행
- 결과값을 HTML로 구성하여 클라이언트에 응답

- HTTP Request(요청)
  - javax.servlet.http.HttpServletRequest
- HTTP Response(응답)
  - javax.servlet.http.HttpServletResponse

```
@WebServlet(name="MyServlet", urlPatterns={"/xxx", "/yyy" })
public class MyServlet extends HttpServlet {

  protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws IOException, ServletException {
     …
  }
}
```

HttpServletRequest API

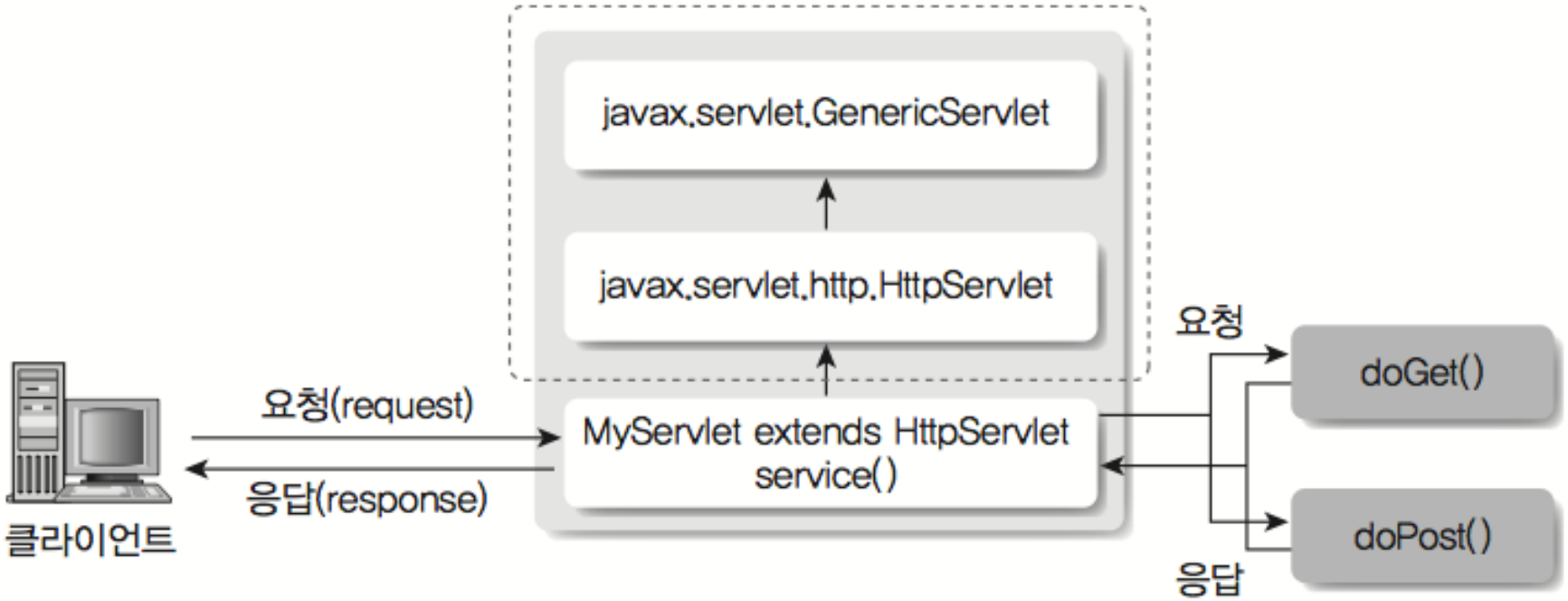
- getHeader(String name):String
- getHeaderNames( ):Enumeration
- getCookies( ):Cookie[ ]
- getRequestURI( ):String
- getServletPath( ):String
- getSession(boolean):HttpSession
- getSession( ):HttpSession
- setCharacterEncoding(String encoding)
- setAttriute(String name, Object obj)
- getAttribute(String name):Object
- removeAttribute(String name)
- getParameter(String name)
- getParameterNames( ):Enumeration
- getParameterValues(String name):String[ ]

HttpServletResponse

- addCookie(Cookie c)
- addHeader(String name, String value)
- encodeURL(String url)
- getStatus( )
- sendRedirect(String loc)
- getWriter( ):PrintWriter
- getOutputStream( ):ServletOutputStream
- setContentType(String type)

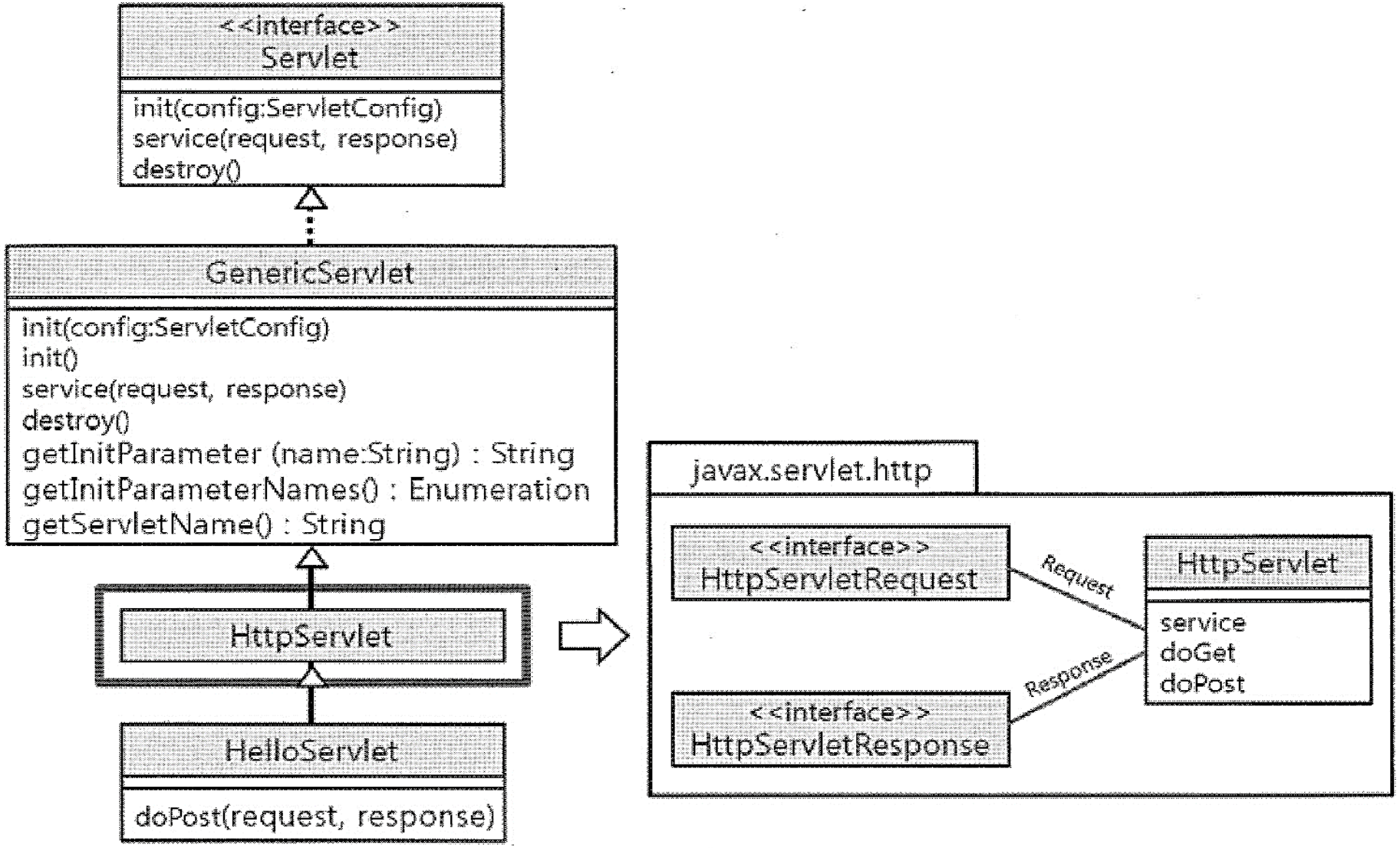
서블릿 응답 처리

- javax.servlet.http.HttpServlet을 상속받은 서블릿 동작 구조



서블릿 아키텍처 및 핵심 API

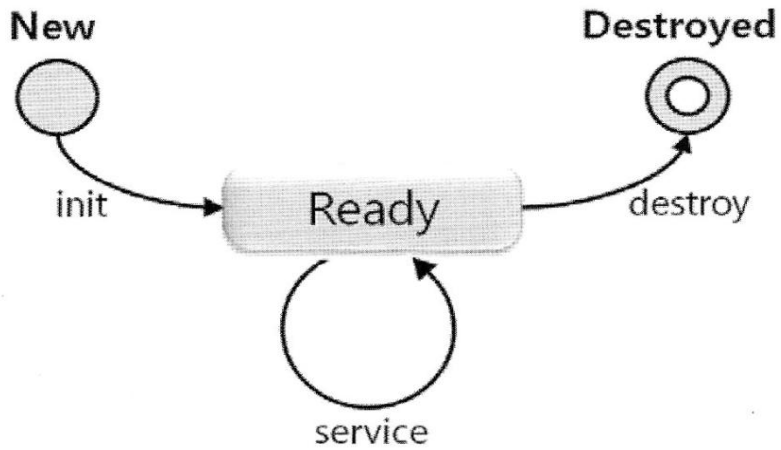
HttpServlet API



서블릿 LifeCycle 메서드

서블릿의 인스턴스를 init, service, destroy 메서드로 관리

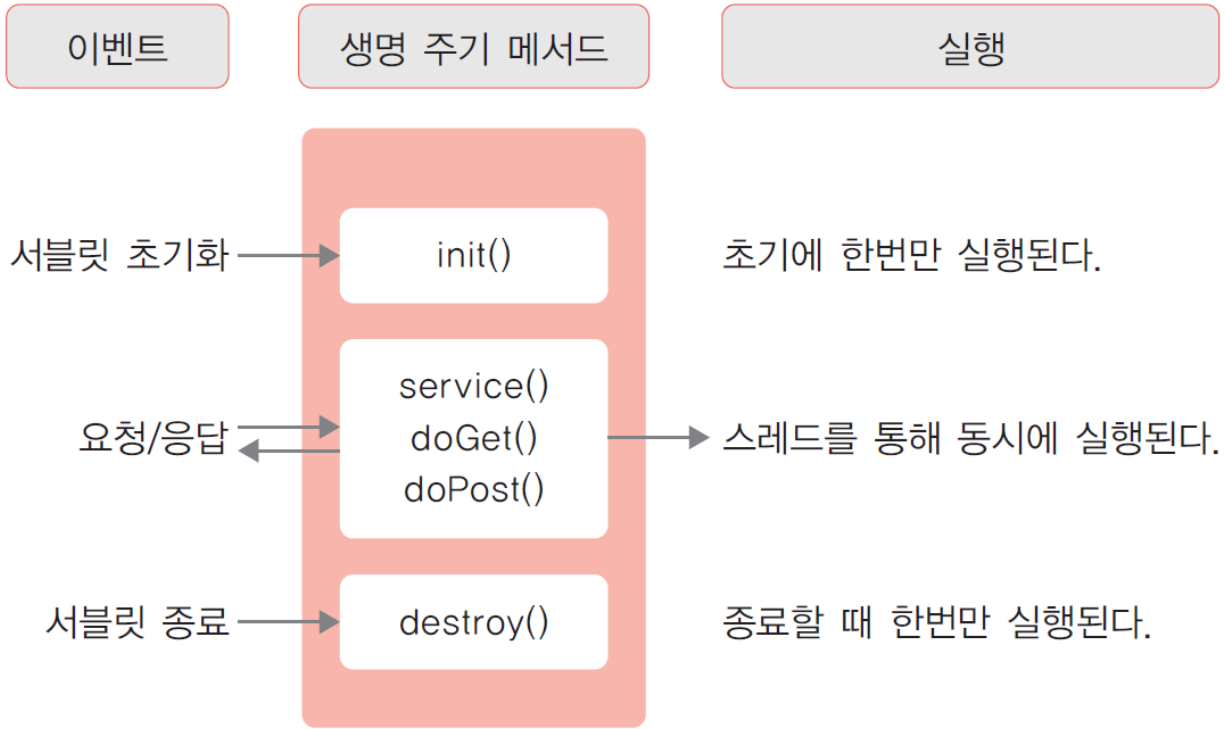
- **init 메서드**
  - 인스턴스가 처음 실행될 때, 단 한번 호출
  - 서블릿에서 필요한 초기화 작업 시 사용
- **service 메서드**
  - 클라이언트가 요청할 때마다 호출
  - doGet 또는 doPost에서 주로 작업
- **destroy 메서드**
  - 인스턴스가 웹 컨테이너에서 제거될 때 호출



서블릿 LifeCycle 메서드

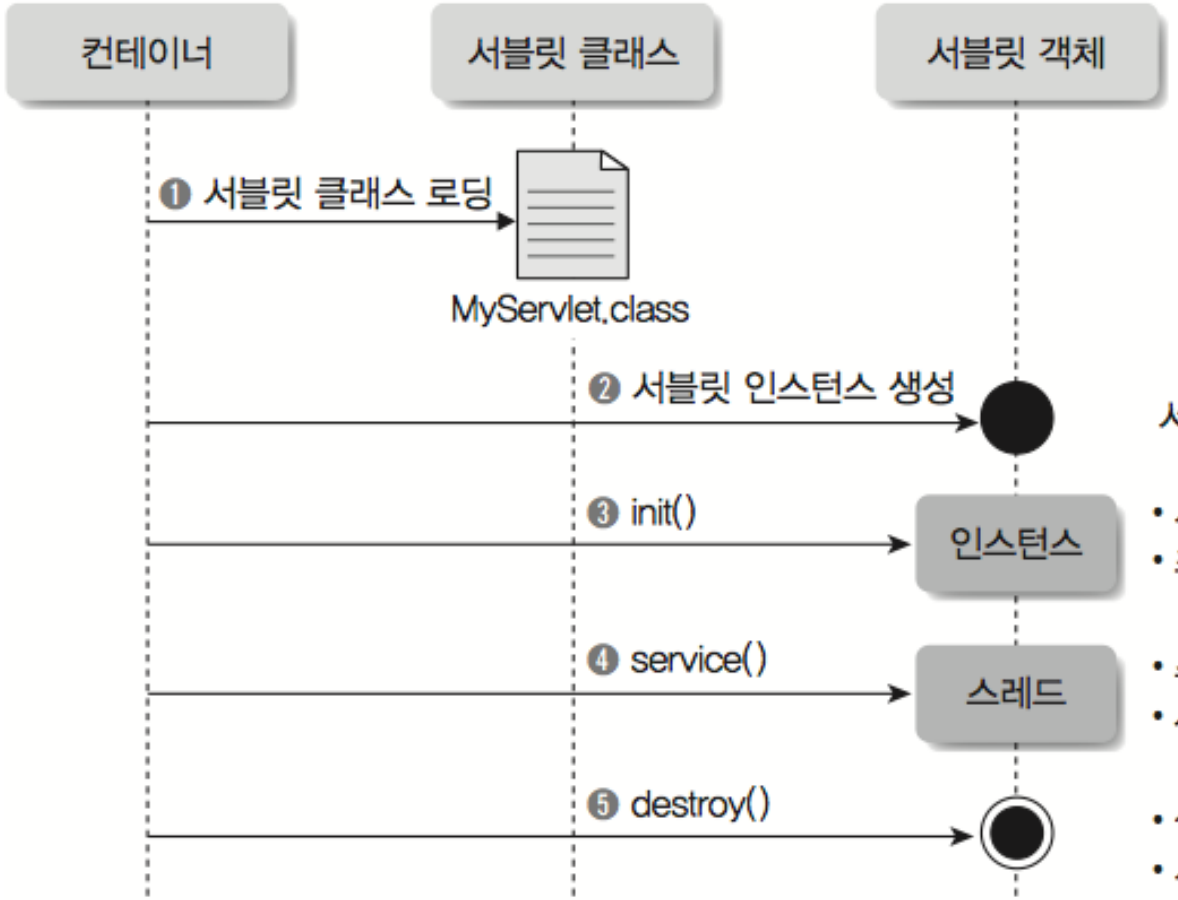
서블릿의 생명 주기

- 객체의 생성에서 종료에 이르는 과정



서블릿 LifeCycle 메서드

서블릿의 동작 과정



서블릿 생성자가 호출된다.

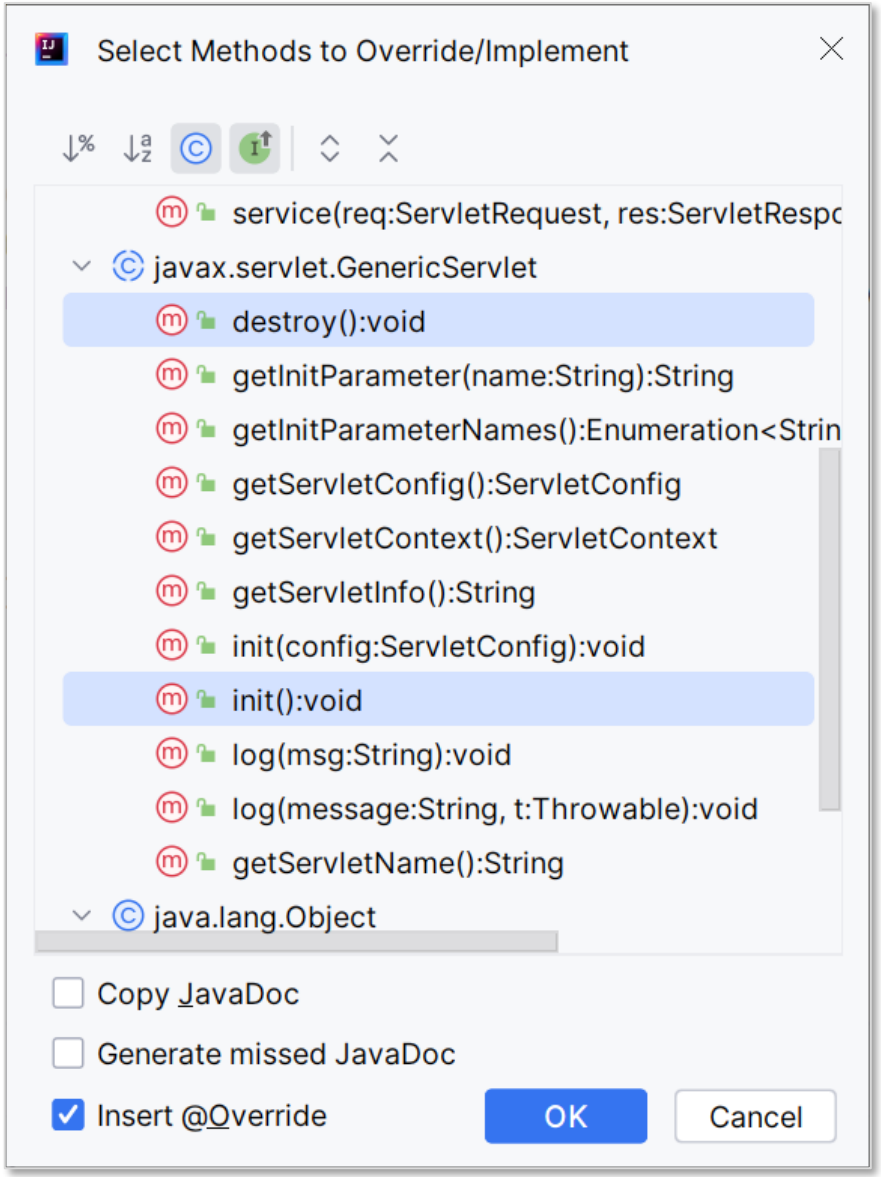
- 서블릿 클래스에서 단 한 번만 실행되는 메서드다.
- 초기화 작업을 수행한다.

- 스레드 사용자 요청을 처리한다.
- 사용자 요청에 따라 service() 메서드가 반복해서 호출된다.

- 상태 종료될 때 호출되는 메서드다.
- 서블릿이 소멸된다.

서블릿 LifeCycle 메서드

- **init, destroy 메서드 추가**
  - HelloServlet.java에서 Ctrl+O
  - init(): void, destroy(): void 선택



서블릿 LifeCycle 메서드

MyServlet init, destroy 메서드 추가

```
@WebServlet(name="MyServlet", urlPatterns={"/xxx", "/yyy" })
public class MyServlet extends HttpServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws IOException, ServletException {
        …
    }

    @Override
    public void destroy() {
        System.out.println("destroy 호출");
    }

    @Override
    public void init() throws ServletException {
        System.out.println("init호출");
    }
}
```

서블릿 응답 처리

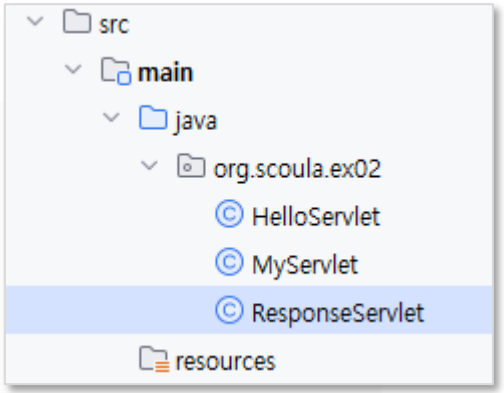
- 클라이언트에서 서블릿으로 요청
- 서블릿은 처리한 결과를 html 형식으로 응답 처리

- 문자셋 설정
  - response.setContentType("text/html;charset=UTF-8");
    - 응답 데이터의 MIME 타입

- 응답 데이터의 전송
  - 문자 데이터 응답
    - response.getWriter()로 PrintWriter 클래스 사용
  - 바이너리 데이터 응답
    - response.getOutputStream()으로 ServletOutputStream 클래스 사용

서블릿 응답 처리

ResponseServlet.java




```
@WebServlet("/response")
public class ResponseServlet extends HttpServlet {

  protected void doGet(HttpServletRequest req, HttpServletResponse resp)
                throws IOException, ServletException {

    // MIME 타입 설정
    resp.setContentType("text/html;charset=UTF-8");

    // 자바 I/O
    PrintWriter out = resp.getWriter();

    // html 작성 및 출력
    out.print("<html><body>");
    out.print("ResponseServlet 요청 성공");
    out.print("</body></html>");

  }
}
```

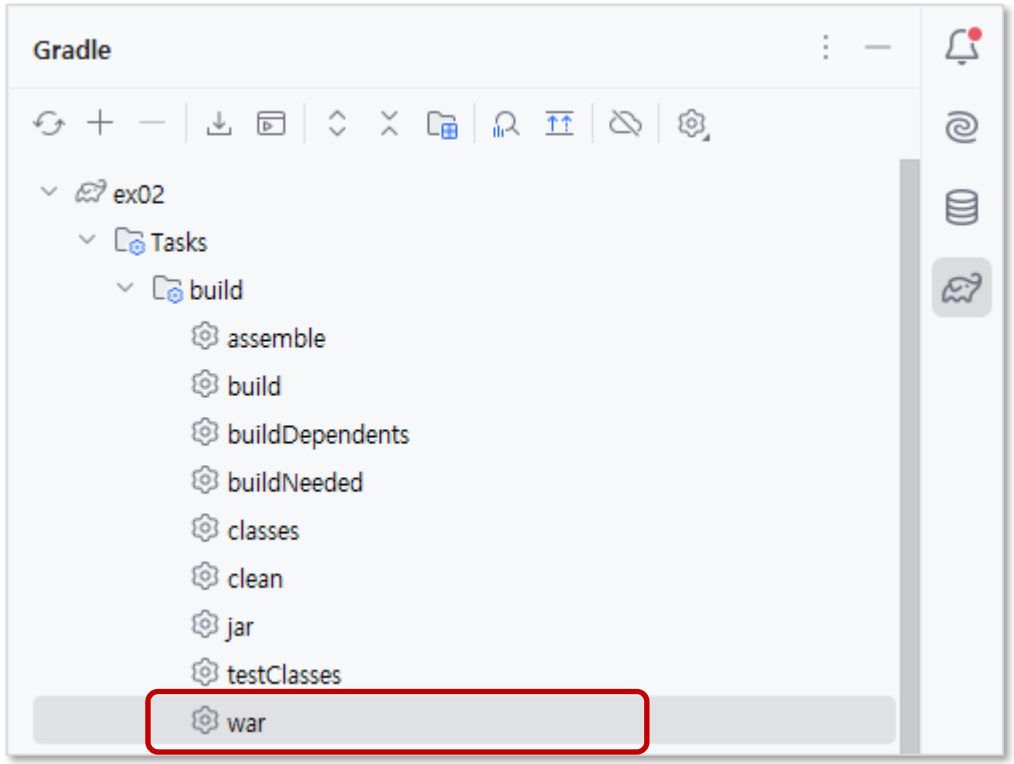
http://localhost:8080/response

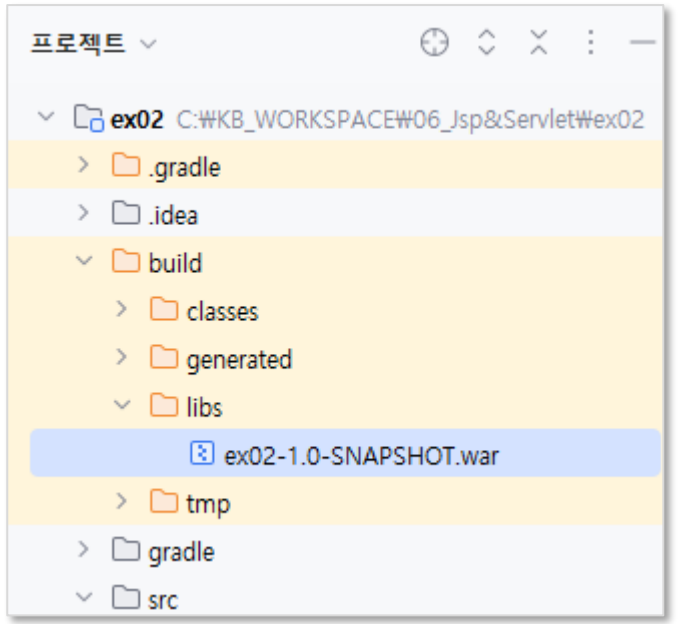
ResponseServlet 요청 성공

ResponseServlet ?? ??

배포하기

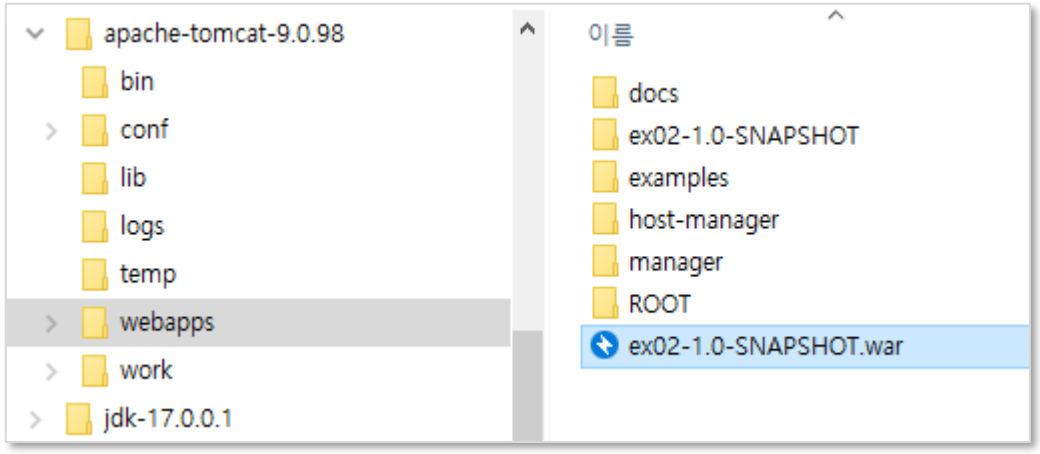
war 파일 만들기





배포하기

war 파일 배치



배포하기

ROOT 애플리케이션 설정

C:/apache-tomcat-9.0.98/conf/server.xml

```
    …
      <Host name="localhost"  appBase="webapps" unpackWARs="true" autoDeploy="true">
        …
        <Valve className="org.apache.catalina.valves.AccessLogValve" directory="logs"
               prefix="localhost_access_log" suffix=".txt"
               pattern="%h %l %u %t &quot;%r&quot; %s %b" />

        <Context path="/" docBase="ex02-1.0-SNAPSHOT"  reloadable="false" > </Context>
      </Host>
    </Engine>
  </Service>
</Server>
```

배포하기

실제 Tocamcat 서버 실행

- C:₩apache-tomcat-9.0.98₩bin₩startup.bat 실행
  - 윈도우인 경우 기동 될 때 콘솔에 출력되는 한글 깨짐
  - http://localhost:8080/ 접속

Tomcat 서버 설치

콘솔 한글 로그 문자셋 설정(윈도우 만)

- C:₩apache-tomcat-9.0.89₩conf₩logging.properties

```
...
############################################################
# Handler specific properties.
# Describes specific configuration info for Handlers.
############################################################

…
java.util.logging.ConsoleHandler.encoding = EUC-KR

...
```

→ IntelliJ에서 실행하는 경우 한글 깨짐!!